

Varun Sharma

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Data Analyst with 3+ years of experience building AI/ML solutions and full-stack systems delivering up to 70% efficiency gains. Skilled in Python, SQL, JavaScript/TypeScript, and Java for predictive modeling, statistical analysis, and LLM integration. Proven expertise in scalable data pipelines, ETL workflows, real-time dashboards, and production-grade ML on AWS/Azure. Specialized in RAG systems, A/B testing, and transforming complex data into strategic insights that shorten development cycles and enhance decision-making.

EDUCATION

Drexel University

Master of Science in Data Science, GPA: 3.91/4.0

Coursework: Machine Learning, Deep Learning, Statistical Analysis, Database Management, Applied Analytics.

Sep 2022 – Jun 2024

Philadelphia, USA

SRM Institute of Science and Technology

Bachelor of Technology in Computer Science & Engineering, CGPA: 6.88/10.0

Coursework: Data Structures, Algorithms, Database Management Systems, Software Engineering.

Jul 2016 – May 2020

Chennai, India

SKILLS

Programming Languages: Python (Advanced), Java (Core Java, Java EE), JavaScript/TypeScript, C#, SQL, C++, .NET Framework, VB/ASP.NET, LLM Function Calling, Prompt Engineering.

Frameworks & Tools: Spring (Framework & Boot), FastAPI, Next.js, React, LangChain, HuggingFace, Node.js, .NET Core.

Machine Learning: RAG Systems, Vector Embeddings, Transformer Models, Predictive Modeling, Statistical Analysis.

Cloud & Infrastructure: AWS, Azure, OpenShift, Docker, Apache Airflow, RESTful APIs, Microservices Architecture

Data Visualization & BI Tools: Tableau, Power BI, Streamlit.

Databases: PostgreSQL, MySQL, MongoDB, Vector Databases, Data Warehousing, ETL Processes, Database Design & Optimization.

DevOps & Development Practices: Git, CI/CD, Performance Optimization, System Debugging, Load Balancing, Object-Oriented Design Patterns, Software Development Lifecycle (SDLC), Agile Methodologies.

PROFESSIONAL EXPERIENCE

AI Intern | Smart Cookie Rewards Pvt. Ltd. | New York, USA

Jan 2025 - Present

- Built production-grade RAG chatbot using LangChain, HuggingFace and Groq LLaMA-3.1 with **85%** query accuracy and support for concurrent real-time chats.
- Designed a full-stack ML pipeline with automated text processing and real-time inference, cutting manual search time by **70%**.
- Created a responsive Streamlit interface with real-time chat and source validation, boosting user engagement by **60%**.
- Improved performance with caching and error handling, reducing API latency by **45%**.

Data Analyst | Drexel University | Philadelphia, USA

Oct 2024 - Present

- Designed automated data pipelines and real-time reporting systems using Python and SQL, improving forecast accuracy by **22%**, query latency by **40%**, and operational efficiency by **30%**.
- Built CLIP-powered dashboards with TensorFlow and PyTorch for multimodal data, boosting strategic decisions by **25%**.
- Optimized data infrastructure and resolved bottlenecks, maintaining **98%+** data quality via proactive monitoring.

Data Analyst Intern | Drexel University | Philadelphia, USA

Sep 2023 - Dec 2023

- Engineered end-to-end analytics solutions in Python with automated validation, accelerating insights delivery by **45%**.
- Built robust ETL pipelines integrating multi-source data, boosting organizational data accessibility by **40%** and maintaining **98%+** data quality using TensorFlow and PyTorch.
- Streamlined reporting workflows to deliver timely business insights, reducing turnaround time by **35%** for strategic decisions.

Software Engineer | Cognizant Technology Solutions | Bangalore, India

Jul 2020 - Jul 2022

- Developed enterprise applications with Core Java, Spring, and React, using automated testing to accelerate dev cycles by **40%** and ensure **99.9%** uptime for critical systems.
- Designed cloud-native solutions on Azure, AWS, and OpenShift with CI/CD automation, cutting provisioning time by **60%** and scaling **15+** client apps.
- Drove SDLC with OOP patterns and automation, cutting defects by **35%** and time-to-production by **50%**.
- Streamlined offshore collaboration with automated tracking, boosting delivery by **30%** and cutting delays by **45%**.
- Led microservices adoption and tech evaluations, boosting performance by **25%** and cutting maintenance by **40%**.

PROJECTS

AI Trading Risk Management | S&P 500 Prediction System

Jun 2025

- Built RNN-based AI trading system (**90.4%** accuracy), cutting risk by **20%** and improving efficiency by **25%**.
- Automated feature pipeline on **497K+** records, generating **17** indicators and boosting model performance by **43%**.
- Developed real-time risk tool (**9.6%** MAPE) for institutional trading and data-driven decision-making.